



CALIBER 2026 — CASE 2: INTELLIGENT MANUFACTURING

One file. One truth. From data to decision.

An offline Intelligent Manufacturing dashboard for Chandra Asri. No server, no internet, no install.

380

incidents, 23 columns

US\$67.19M

total exposure

2,261.1 h

downtime tracked

87.1%

AI retrieval @ rank 1

THE PROBLEM

Six functions. Six dashboards. Zero agreement.

Operations, Production, Energy, Maintenance, HSE, and Management each run different systems, KPI definitions, formats, and timelines.

Slow validation

Hours lost comparing reports before any decision is made.

Reactive decisions

Root causes found late, critical alerts drowned in noise.

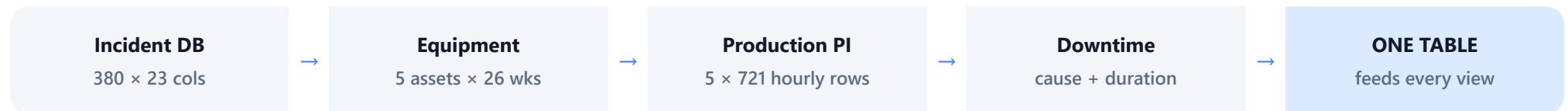
Open loop

Follow-up actions assigned verbally, completion never tracked.

KEY QUESTION 1 — DATA FOUNDATION

Four sources in. One governed table.

Every chart on every tab reads the same incident table. Edit one record, every figure recomputes.



KEY QUESTION 2 — SINGLE PANE OF GLASS

The executive view answers in 5 seconds.

US\$67.19M

hero exposure card

ZCU worst

US\$11.11M · 62 inc

Leakage #1

101 cases, fix first

179

overdue, oldest Feb 24

- 8 KPI cards, unit ranking, monthly trend, fix-first priority — every card clickable to its source rows.
- Trip logsheet: one row per trip, color-coded duration, cause, and follow-up dots.

KEY QUESTION 3 — AI ROOT CAUSE + ACTION

AI suggests. Humans close.

87.1%

similar-incident retrieval @ rank 1 (leave-one-out, n=380)

98.4%

retrieval @ rank 5 — with confidence intervals

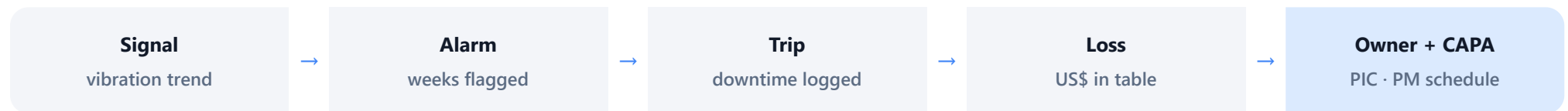
Top 10

exposure plan drafted in one click, owner attached

- Problem tank → root-cause indication → prioritized alerts → tracked action with guidance.

CASE RECONSTRUCTION — COMPRESSOR KO-3201

Signal to owner, end to end.



- All 5 RCA decks integrated (4P / 4M+1E, CAPA, PM schedule, PIC, status).
- Chronology, correlation with same-mechanism records, recommendation — one tap from any row.

CLOSING THE LOOP

Insight becomes action, action becomes evidence.

Alarm dispatch

WhatsApp to PIC or shift group. Every send logged.

Follow-up board

Owner, due date, progress, evidence per action.

6 reports

Print-ready PDF: ops, incidents, RCA, follow-ups, personnel, energy.

IMPACT — STATED HONESTLY

What changes, what we label, what we skip.

- 179 overdue actions surfaced with owner and age — oldest due Feb 2024, none hidden.
- Energy = derived proxy ($P = \sqrt{3} \times 400 \times I \times 0.86 / 1000$), formula shown — never called metered.
- CCTV wall = labeled mock illustration — unit aggregates underneath are real.
- No live streaming (not required) — architecture ready for PI stream when available.

TEAM & NEXT

Jenny • Hafidz • Naufal

Advisor: Mr. Bayu Aji. Built from the official baseline only, plus two declared additions.

Next: meter it

5 metered energy tags to replace the proxy formula

Next: connect it

EDMS / AIMS / Digital Twin integration for asset context

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